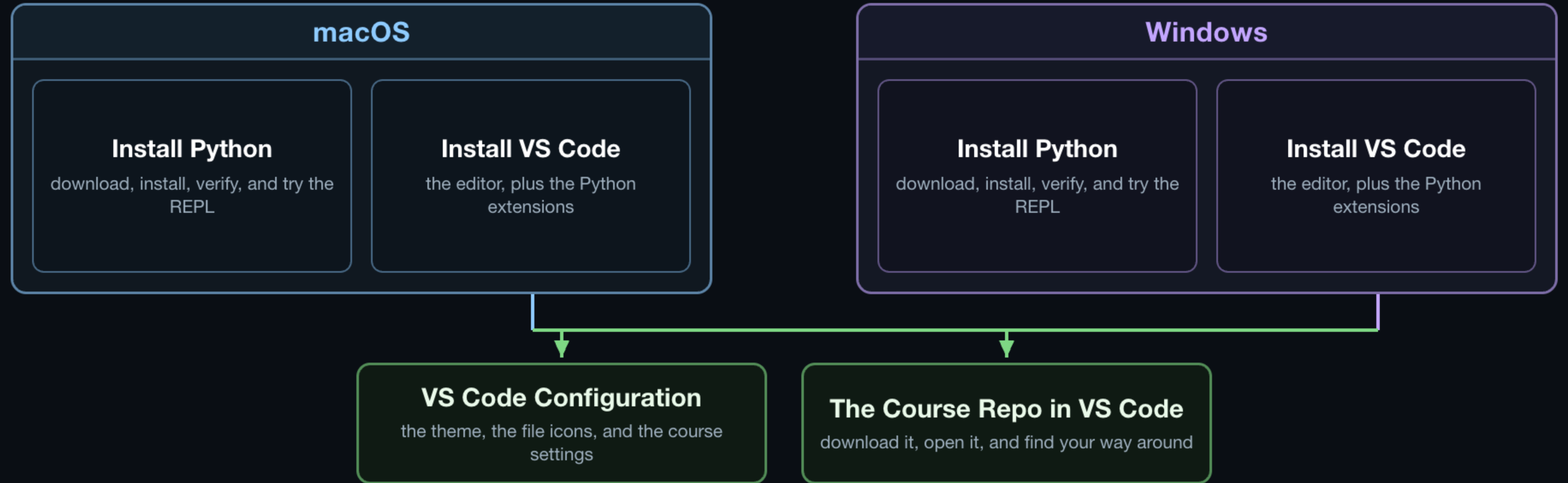


Section 01: Python Environment Setup

Pick your operating system, install two things, then everyone meets again for the same two steps.



WATCH THE VIDEOS FOR YOUR OWN OS

On **macOS**, follow the macOS install videos. On **Windows**, follow the Windows ones. From **VS Code Configuration** onward, every video is the same for everyone.

Setup: Install Python and the REPL

The same three steps on either system. Only the command name changes.

macOS Linux is the same

- 1 **Download** from python.org/downloads, the official site.
- 2 **Install** it: double-click the package and follow the installer.
- 3 **Verify**, then start the shell. macOS ships an older Python, so always say `python3`.

```
$ python3 --version
Python 3.14.6
$ python3
>>> 2 + 2
4
```

Windows Windows Terminal

- 1 **Download** from python.org/downloads, the official site.
- 2 **Install** it, and on the FIRST installer screen:
Tick "Add python.exe to PATH" or Python will not run
- 3 **Verify**, then start the shell. On Windows the command is `python`, with no 3.

```
> python --version
Python 3.14.6
> python
>>> 2 + 2
4
```

That `>>>` is the REPL: it runs one line at a time and answers straight away.

Setup: What We Will Do in VS Code

Three things, in this order. Then you are ready to write Python.

1 Install VS Code

Download it, install it, and open it until the **Welcome** tab appears.

- code.visualstudio.com
- the .dmg, or the Windows installer
- the Welcome tab

This step differs by operating system

2 Add four extensions

Python support, a fast linter, a dark theme, and clear file icons.

- Python
- Ruff
- One Dark Pro
- Material Icon Theme

Same on every operating system

3 Change three settings

Roomier folder indents, line wrapping, and **auto save**, so you never lose a change.

- `tree indent`
- `word wrap`
- `auto save`

Same settings, only the keys differ

Do this once now, and the editor stays set up for the whole course.

Setup: Install VS Code

The editor you will write every line of this course in. Install it once.

macOS Linux is similar

- 1 **Download** from code.visualstudio.com.
- 2 **Open the .dmg** and drag Visual Studio Code into **Applications**.
- 3 **Launch it**: press **Cmd + Space**, type Visual Studio Code, press Enter.

You are done when the Welcome tab opens

Windows the installer does more

- 1 **Download** from code.visualstudio.com.
- 2 **Run the installer** and tick **Add to PATH** and both **Open with Code** options.
- 3 **Launch it** from the **Start menu**.

You are done when the Welcome tab opens

Setup: Install the VS Code **Extensions**

Extensions icon in the left sidebar, search the name, click Install. Check the publisher.

INSTALL THESE FOUR YOURSELF

Python Microsoft

Run and debug code, pick your interpreter, autocomplete. The must-have one.

Ruff Astral Software

A very fast linter. It underlines mistakes as you type.

One Dark Pro binaryify

A clean dark colour theme that is comfortable to read.

Material Icon Theme Philipp Kief

Coloured icons per file type, so you spot a .py at a glance.

PYTHON BRINGS THESE ALONG

Pylance

The engine behind autocomplete, type hints and go to definition.

Python Debugger

Set breakpoints and step through your code to find bugs.

Python Environments

Create and manage virtual environments and packages from the UI.

You do not install these three. They arrive with the Python extension.

COMMAND PALETTE

macOS

Cmd + Shift + P

Windows

Ctrl + Shift + P

How you run any command in VS Code

THEN TURN THEM ON

Installing does not switch them on. Open the Command Palette above, then pick **Preferences: Color Theme** and **Preferences: File Icon Theme**.

Setup: Update Important VS Code Settings

Only the key to open Settings differs. The three changes are the same everywhere.

macOS Linux is similar

Cmd + , open Settings

Windows

Ctrl + , open Settings

THEN TYPE EACH ONE IN THE SEARCH BOX AND SET THE VALUE

SEARCH FOR

SET IT TO

WHY

tree indent

20

More space between nested folders, so the project is easy to read.

word wrap

on

Long lines wrap instead of scrolling off the right edge.

auto save

afterDelay

Off by default, so this one really matters. Saves your file one second after you stop typing.

Setup: Get the Course **Code Repo**

Download it once. From section 02 on, every section folder looks the same inside.

COURSE CODE REPO

<https://github.com/stacksimplify/python-for-absolute-beginners>

BOOKMARK IT

1 Download

Open the link above, click the green **Code** button, then **Download ZIP**.

2 Unzip

Put it in **Documents**, not Downloads.

macOS double-click

Windows Extract All

3 Rename

Drop the **-main** branch suffix. See below.

4 Open

In VS Code use **File** then **Open Folder**. Open the **folder**, never a single file.

RENAME ~~python-for-absolute-beginners-main~~ → **python-for-absolute-beginners**

-main is the branch name, not part of the project

WHAT YOUR EXPLORER SHOULD SHOW

python-for-absolute-beginners/

01-Python-Environment-Setup **README only, no code**

02-Variables-and-Data-Types

... one folder per section ...

13-Classes-and-OOP-Basics

milestone-projects the projects you build

pdfs section slides to keep

images you will not open this

INSIDE EVERY SECTION FROM 02 ON

05-Collections/

python-learn concept demos

s05_lecture.py you type along here

python-practice

solutions

python-workshop

solutions

12 is command line work, so it has no python-workshop

Leave this folder open in VS Code for the whole course. Every section you start is a folder you already have.